



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

Experiment No.8
To use bulk_extractor tool to detect Evidence related to email
Date of Performance:
Date of Submission:



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Aim: To use bulk_extractor tool to detect Evidence related to email

Objective: To make use of the bulk_extractor tool to recover email related evidence from a disk image

Theory:

Bulk_extractor is a C++ program that scans a disk image, a file, or a directory of files and extracts useful information without parsing the file system or file system structures. The results are stored in feature files that can be easily inspected, parsed, or processed with automated tools. bulk_extractor also creates histograms of features that it finds, as features that are more common tend to be more important.

Bulk Email Extractor is also an email client manager. The software provides a streamlined extraction of email. It supports every online email service on the internet. This helps users process all email addresses in an efficient and automated manner.

Bulk Email Extractor is created to handle tons of emails addresses. Collecting email addresses is a monumental job for anyone. Piled-up messages from clients and accounts can be a good source of email addresses. Some of them could be not existing or fake and most users don't have the luxury of checking individual email addresses. It will take several hours until all email addresses are collected and sorted out. This software is capable of extracting large amounts of email addresses. It runs within minutes compared to the manual opening of contact addresses. Any online email services or business directory websites are supported by this software and it even supports websites with email listings.

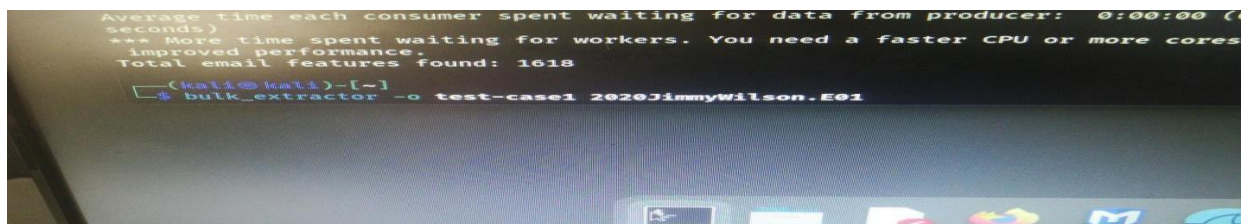


Fig. 8.1 View of Bulk_Extractor

Process:

Step 1, open the Bulk_extractor tool in kali Linux

Step 2. create or download the mirror image of the hard disk



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Step 3. use the command as shown in figure 8.1
Step 4. open the output file to see email evidence

Output:

```
root@kali: /home/kali
bulk_extractor -o test-case1 ntfs1-gen2.E01
mkdir "test-case1"
opening ntfs1-gen2.E01

bulk_extractor version: 2.1.1
Input file: "ntfs1-gen2.E01"
Output directory: "test-case1"
Disk Size: 516554752
Scanners: aes base64 elf evtx exif facebook find gzip httplogs json kml_carved msxml net ntfsindx ntfslogfile ntfsmft ntfsusr pdf rar sqlite utmp vcard_carved windirs w
inlnk winpe winprefetch zip accts email gps
Threads: 2
going multi-threaded...( 2 )
bulk_extractor Mon Mar 24 02:43:54 2025

available_memory: 4921139200
bytes_queued: 0
depth0_bytes_queued: 0
depth0_sbufs_queued: 0
elapsed_time: 0:00:00
estimated_date_completion: 2025-03-24 02:43:53
estimated_time_remaining: n/a
fraction_read: 0.000000 %
max_offset: 0
sbufs_created: 1
sbufs_queued: 1
sbufs_remaining: 1
tasks_queued: 0
thread_count: 2
>.....|

bulk_extractor Mon Mar 24 02:43:55 2025

available_memory: 4874383360
bytes_queued: 104857600
depth0_bytes_queued: 104857600
depth0_sbufs_queued: 5
```

```
bulk_extractor Mon Mar 24 02:44:16 2025

available_memory: 4823572480
bytes_queued: 52953088
depth0_bytes_queued: 52953088
depth0_sbufs_queued: 4
elapsed_time: 0:00:22
estimated_date_completion: 2025-03-24 02:44:16
estimated_time_remaining: 0:00:00
fraction_read: 100.000000 %
max_offset: 503316480
sbufs_created: 866936
sbufs_queued: 4
sbufs_remaining: 1
tasks_queued: 2
thread-1: 503316480: accts (13238272 bytes)
thread-2: 503316480: net (13238272 bytes)
thread_count: 2
=====|

Phase 2. Shutting down scanners
Computing final histograms and shutting down...
Phase 3. Generating stats and printing final usage information
All Threads Finished!
Elapsed time: 23.19 sec.
Total MB processed: 516
Overall performance: 22.28 MBytes/sec 11.14 (MBytes/sec/thread)
sbufs created: 866936
sbufs unaccounted: 0
Time producer spent waiting for scanners to process data: 0:00:16 (16.69 seconds)
Time consumer scanners spent waiting for data from producer: 0:00:00 (0.63 seconds)
Average time each consumer spent waiting for data from producer: 0:00:00 (0.00 seconds)
*** More time spent waiting for workers. You need a faster CPU or more cores for improved performance.
Total email features found: 9

root@kali: /home/kali
```



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```
# BANNER FILE NOT PROVIDED (-b option)
# BULK_EXTRACTOR-Version: 2.1.1
# Feature-Recorder: email
# Filename: ntfs1-gen2.E01
# Feature-File-Version: 1.1
47346547      xL6@3ov.KE      ^g\214\007t\344g'DF\230\230\364F\264\204xL6@3ov.KE\343\341#\275\231\014\357e\213su\226\203\350e\227
239767510-PDF-392      btinfo@bottomline.com      .bottomline.com      btinfo@bottomline.com      Paybase Allows
239767510-PDF-1681      info@entegrity.com      w.entegrity.com      info@entegrity.com      Equifax E-Banki
239772263-PDF-1084      paul.wrenn@hicor.net      4 www.hicor.net      paul.wrenn@hicor.net      Mellon Global C
239772263-PDF-1296      gcm_direct_pgh@mellon.com      n.com/inst/gcm/      gcm_direct_pgh@mellon.com      National City C
239772263-PDF-1673      mark_d_schulte@national-city.com      14 216-222-3633      mark_d_schulte@national-city.com      \134(continued\134
239776426-PDF-437      gigiw@paytec.com      55 717-506-2200      gigiw@paytec.com      Politzer & Hane
239776426-PDF-1747      ed_armstrong@stercomm.com      ingcommerce.com      ed_armstrong@stercomm.com      \134(continued\134
239780882-PDF-812      patricia.engelage@umb.com      045 www.umb.com      patricia.engelage@umb.com
```

Conclusion:

In this experiment, the Bulk_Extractor tool was effectively utilized to extract digital artifacts from large data sources without parsing entire file systems. Its capability to scan disk images and identify critical information such as email addresses, credit card data, and URLs makes it an essential tool in digital forensics. By automating the process of artifact extraction, Bulk_Extractor accelerates evidence analysis and enables investigators to uncover hidden or suspicious patterns with ease. This makes it highly valuable in cases where timely discovery of relevant digital traces is crucial for solving cybercrime investigations.